Closed-loop model of the LUCY mount: identification and validation on the October 2026 tests

Generated by `build_vcl_report.py`
`sim/gimbal_multidof/closedLoopModel/report_vcl`

October 20, 2026

Purpose. This report traces the work done on the tests of October 2026 (folder `LucyIdentification/VClosedLoop`, used as the only source of truth): what was done at each iteration, what changed from the previous one and why, and which results were obtained. It ends with the role of the parameters, the identifiability analysis of the fitted models and the next steps.

Two parts.

- **Part A, closed-loop study** (`vcl_study.py`). The model includes the drives:
  - transfer functions of the two controllers;
  - fit of the plant on the frequency responses;
  - validation of the whole loop on the step sequences, with five iterations on the controller (I1–I5).
- **Part B, current-driven study** (`current_input_study.py`). The measured motor current is the *input* of the model, so the drives are not in the loop, and the only measure of goodness is the fit of the encoder position. There are six runs: first run, I6, I7, I8, I9a, I9b.

Main results.

- **Controller.** The transfer functions built from the drive files reproduce the WorkBench Controller and Feedback Biquad traces to 0.001 dB and 0.25°.
- **Closed loop.**
  - The drive files reproduce the step tests only for the azimuth, and only with `PL.KP` read in 1/s.
  - With identified effective gains (I5), the position-error fit on unseen moves is 83 % for the azimuth and 81 % for the elevation.
  - The current fit is 45 % for the azimuth and 68 % for the elevation.
- **Current-driven plant.**
  - The azimuth position is reproduced on unseen moves with a fit of 94–95 %.
  - The elevation stays at 86–87 %, below the 90 % threshold.
  - Neither a two-level hysteresis, nor a trigonometric position torque, nor a cable spring with one neutral angle generalises.
  - The identifiability analysis shows the reason: the 1° step sequences hold only a few tens of independent samples per axis. The friction parameters are therefore poorly determined and strongly correlated, and adding parameters increases the overfitting.

Contents

1 Data and model **2**

2 Part A: closed-loop study **2**
2.1 Steps 1–2: controller transfer functions 2
2.2 Steps 3–5: plant parameters on the frequency responses 4
2.3 Step 6: validation of the closed loop, iterations I1–I5 5
2.4 Imbalance of the telescope 6

3 Part B: plant driven by the measured current **7**
3.1 Method 7
3.2 Runs and what changed 8

4 Role of the parameters **11**

5 Identifiability analysis **11**

6 Next steps **13**

A Configurations and results **13**

1 Data and model

Data (VClosedLoop).

- drive parameter files `Parameters_azimuth.akd` and `Parameters_elevation.akd`;
- WorkBench frequency responses `BodeMeasurement_{azimuth,elevation}.csv`: current excitation 3.5 and 3 A, 1–2000 Hz, measured plant together with the traces computed by WorkBench;
- four step sequences `MPCLucy_*.csv`, sampled at 2 kHz:
  - 1 and 2° moves of each axis, trapezoidal speed profiles at 10 deg/s;
  - channels: command and encoder position in millidegrees, current `AXIS*_AMPS` (2 units per A_{rms}), torque estimated by the drives.

Split of the sequences. The 1° sequences (`MPCLucy_step_1deg`, `MPCLucy_1degAz_zeroAz_1degPol_zeroPol`) are used for identification. The 2° sequences (`MPCLucy_2degAz_2degPol_zeroAz_zeroPol`, `MPCLucy_2degPol_zeroPol_2degAz_zeroAz`) are used for validation.

Model. The closed-loop library `lucy_cl` has three components:

- the nonlinear 8-DOF lumped plant H6 of `gimbal_mdof`, with $\mathbf{q} = [\psi_e, \psi_m, \psi_L, \theta_e, \theta_h, \theta_m, \theta_r, \theta_p]$;
- the TBM(S)-7646-A motors;
- the AKD2G drives: position P, series velocity PI, biquads AR1–AR4, observer, current PI.

The telescope is balanced ($g = 0$) unless stated otherwise. The time simulations start from the measured state at rest before each move, are driven by the measured commands (Part A) or currents (Part B), and are compared through the NRMSE fit of the System Identification Toolbox,

$$\text{fit} = 100\Big(1 - \frac{\|y - \hat{y}\|}{\|y - \bar{y}\|}\Big)\ \%. \tag{1.1}$$

2 Part A: closed-loop study

2.1 Steps 1–2: controller transfer functions

Structure. From the parameter files and the structure of the AKD2G loops,

$$C_p = K_{pp}, \quad C_v = K_{pv}\frac{s + 2\pi K_{iv}}{s}, \quad H_{fwd} = H_{AR1}H_{AR2}, \quad H_{fb} = H_{AR3}H_{AR4}\frac{\omega_o}{s + \omega_o}, \quad C_i = K_{pi}\frac{s + R/L}{s}, \tag{2.1}$$

with the biquads in the continuous form

$$H(s) = \frac{s^2/\omega_z^2 + s/(Q_z\omega_z) + 1}{s^2/\omega_p^2 + s/(Q_p\omega_p) + 1}. \tag{2.2}$$

The control law is

$$I_{cmd} = C_v H_{fwd}\big[C_p(P_{cmd} - P_{fb}) - H_{fb}\, sP_{fb}\big]. \tag{2.3}$$

From it follow:

- the controller from the position error, $C_e = C_v H_{fwd} C_p$;
- the controller from the encoder position, $C_{fb} = C_v H_{fwd}(C_p + sH_{fb})$;
- the WorkBench Controller trace, $C_{WB} = C_v H_{fwd}(1 + 2\pi\,\texttt{PL.KP}/s)$.

Tables 2–3 give order, high-frequency gain (ratio of the leading coefficients), zeros and poles of every block, with $K_{pp} = 2\pi\,\texttt{PL.KP}$. The model reproduces the traces stored in the Bode exports (Table 1, Figure 1).

Units of the files. The two files use different units: azimuth positions in degrees (`UNIT.PROTARY` 2), elevation positions in millidegrees (custom units, `UNIT.PIN` 360000). The gains of the loops do not depend on them.

Table 1: Largest difference between the model and the WorkBench traces of the October Bode exports, 1–2000 Hz.

axis	Controller [dB]	Controller [deg]	Feedback Biquad [dB]	Feedback Biquad [deg]
azimuth	0.0008	0.247	0.0000	0.00
elevation	0.0000	0.053	0.0258	18.43

Table 2: Azimuth controller: order of numerator/denominator, high-frequency gain [A/rad, A/(rad/s), V/A or –], zeros and poles (frequency and damping ratio).

block	order	gain	zeros	poles
C_p	0/0	62.83	–	–
C_v	1/1	250	1 Hz (real)	0 (integrator)
H_{fwd}	4/4	0.000784	65 Hz, ζ 0.0889; 500 Hz, ζ 0.167	14 Hz, ζ 0.167; 65 Hz, ζ 0.0281
H_{AR34}	0/0	1	–	–
H_{obs}	0/0	1	–	–
C_i	1/1	5.938	97.77 Hz (real)	0 (integrator)
C_e	5/5	12.32	1 Hz (real); 65 Hz, ζ 0.0889; 500 Hz, ζ 0.167	0 (integrator); 14 Hz, ζ 0.167; 65 Hz, ζ 0.0281
C_{fb}	6/5	0.196	1 Hz (real); 10 Hz (real); 65 Hz, ζ 0.0889; 500 Hz, ζ 0.167	0 (integrator); 14 Hz, ζ 0.167; 65 Hz, ζ 0.0281
C_{WB}	6/6	0.196	1 Hz (real); 10 Hz (real); 65 Hz, ζ 0.0889; 500 Hz, ζ 0.167	0 (integrator); 14 Hz, ζ 0.167; 65 Hz, ζ 0.0281

Table 3: Elevation controller, as Table 2.

block	order	gain	zeros	poles
C_p	0/0	2.513	–	–
C_v	1/1	25	15 Hz (real)	0 (integrator)
H_{fwd}	4/4	1	50 Hz, ζ 0.0418; 96 Hz, ζ 0.0418	50 Hz, ζ 0.166; 96 Hz, ζ 0.166
H_{AR34}	4/4	0.0006554	5000 Hz, ζ 0.707	800 Hz, ζ 0.707
H_{obs}	0/1	1257	–	200 Hz (real)
C_i	1/1	5.938	97.77 Hz (real)	0 (integrator)
C_e	5/5	62.83	15 Hz (real); 50 Hz, ζ 0.0418; 96 Hz, ζ 0.0418	0 (integrator); 50 Hz, ζ 0.166; 96 Hz, ζ 0.166
C_{fb}	10/10	83.42	0.3987 Hz (real); 15 Hz (real); 50 Hz, ζ 0.0418; 96 Hz, ζ 0.0418; 3030 Hz, ζ 0.853; 4105 Hz, ζ 0.0215	0 (integrator); 50 Hz, ζ 0.166; 96 Hz, ζ 0.166; 200 Hz (real); 800 Hz, ζ 0.707
C_{WB}	6/6	25	0.4 Hz (real); 15 Hz (real); 50 Hz, ζ 0.0418; 96 Hz, ζ 0.0418	0 (integrator); 50 Hz, ζ 0.166; 96 Hz, ζ 0.166

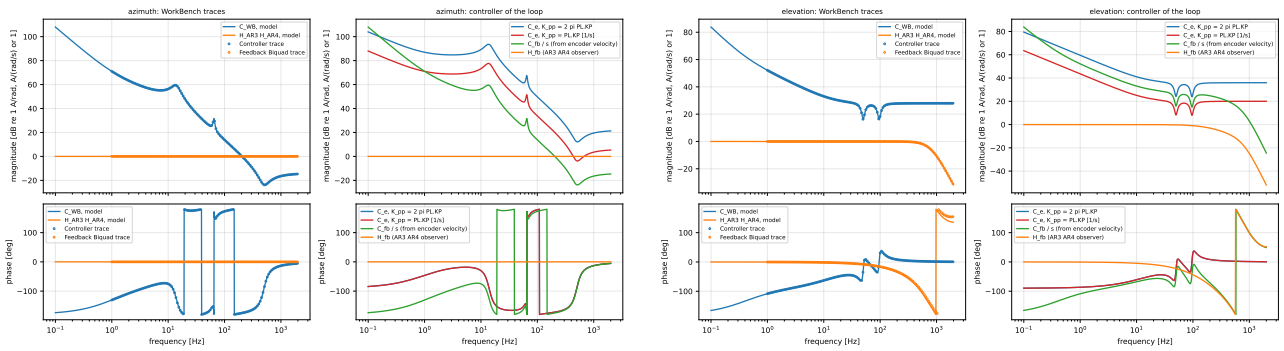


Figure 1: Controllers of azimuth (left) and elevation (right). In each pair, the left panels show the model against the WorkBench Controller and Feedback Biquad traces, and the right panels show the controller of the loop with the two readings of `PL.KP`.

2.2 Steps 3–5: plant parameters on the frequency responses

Parameters. The measured plant (encoder velocity over current command) is fitted with the plant, the motor electrical dynamics and the current loop. The fitted parameters are:

- elevation: the hub–telescope spring $k_b = k_{\theta,4}$, the ring stiffness factor κ_r and the ground spring $k_{g,\theta}(1 + j\eta_\theta)$;
- azimuth: the ground spring $k_{g,\psi}(1 + j\eta_\psi)$;
- both axes: the element dampers.

The ring acts on $k_{\theta,1}$ (encoder–hub) and $k_{\theta,2}$ (hub–motor) as a compliance in series, $1/k = a + b/\kappa_r$, with a and b from two builds of the model (checked exact on other values). It is bounded by $\kappa_r \geq 1$, the ring alone.

Method.

- Residuals $\ln(\hat{G}/G)$, magnitude and phase, robust loss.
- Fit bands where the model has the features of the measurement: elevation 1–200 Hz, azimuth 1–20 Hz for the ground spring and 1–300 Hz for the dampers.
- Three stages: A stiffness and ground, B dampers with a weak prior on their start values, C joint refinement.

Table 4: Fit of the plant on the October frequency responses: stages, band [Hz], fitted values (N m/rad, N m s/rad, –) and fit of the plant [%] per band.

axis	stage	band	fitted parameters	1–5 Hz	5–100 Hz	>100 Hz
elevation	A	1–200	k_b 1.951e5; κ_r 2.726; $k_{g,\theta}$ 188.4; η_θ 2.759	48	27	-44
elevation	B	1–200	$c_{\theta,1}$ 4.964; $c_{\theta,2}$ 0.1592; $c_{\theta,3}$ 0.9451; $c_{\theta,4}$ 24.99	48	27	-117
elevation	C	1–200	k_b 1.972e5; κ_r 1.923; $k_{g,\theta}$ 188.5; η_θ 2.758; $c_{\theta,1}$ 5.04; $c_{\theta,2}$ 0.07975; $c_{\theta,3}$ 0.8987; $c_{\theta,4}$ 24.6	48	27	-123
azimuth	A	1–20	$k_{g,\psi}$ 146.6; η_ψ 1.831	75	5	-97
azimuth	B	1–300	$c_{\psi,1}$ 0.3697; $c_{\psi,2}$ 6.578	75	15	-3
azimuth	C	1–300	$k_{g,\psi}$ 149; η_ψ 1.762; $c_{\psi,1}$ 0.3696; $c_{\psi,2}$ 6.587	75	15	-3

Results.

- **Elevation:** k_b goes from $1.16{\cdot}10^5$ to 1.972e5 N m/rad, the antiresonance from 58 to about 66 Hz; $\kappa_r = 1.92$; ground spring 188 N m/rad with $\eta = 2.76$.
- **Azimuth:** ground spring from 1418 to 149 N m/rad with $\eta = 1.76$; below 5 Hz the plant fit rises from -127 to 75 %.

Limits. Two measured features cannot be reached with these parameters: the elevation resonance at about 95 Hz, and the azimuth resonance at 470 Hz, which the model places at 750 Hz. In both cases the measured level above the antiresonance requires a larger inertia on the motor/hub side than the model has: about 0.26 against 0.034 kg m^2 in elevation, and about 3 times the motor-side inertia in azimuth.

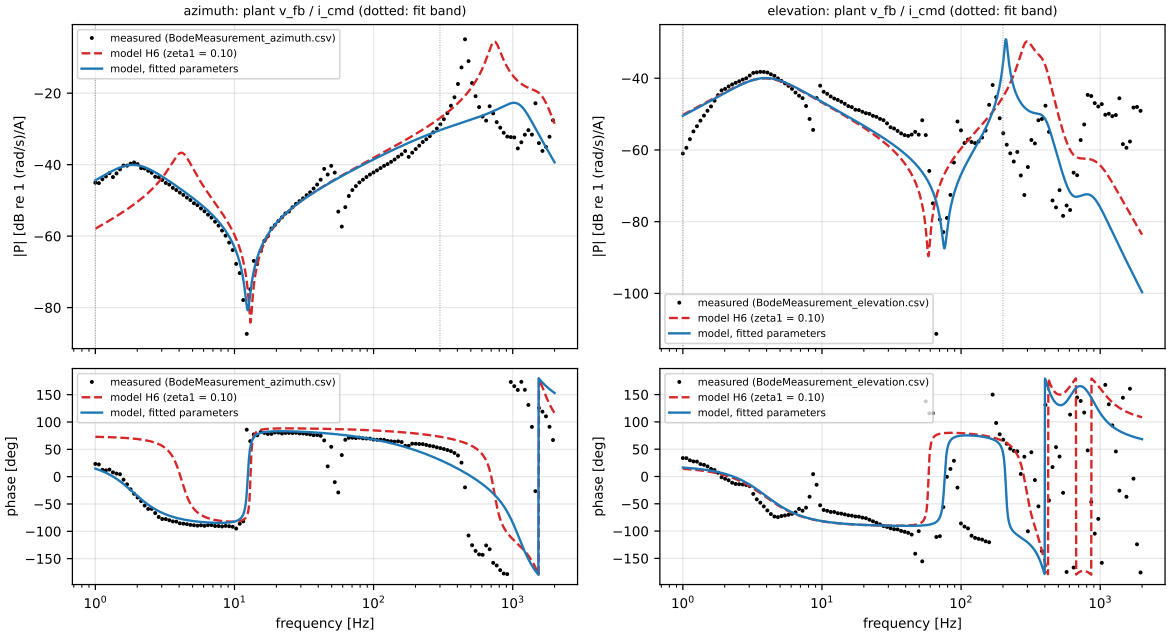


Figure 2: Plant of the October frequency responses: measurement, H6 with $\zeta_1 = 0.10$ and fitted parameters.

2.3 Step 6: validation of the closed loop, iterations I1–I5

Procedure. Every move is simulated with the measured commands of both axes, and the position error and the current of the moving axis are compared. The model is adequate when the median fit is at least 70 % on the error and 50 % on the current, on each axis. When it is not, the procedure goes back to the controller (step 1). Table 5 lists the iterations. I1 and I2 are evaluated on all four sequences; I3–I5 identify on the 1° sequences and validate on the 2° ones.

Table 5: Closed-loop iterations: what changed, effective gains (K_{pp} [1/s], K_{pv} [A/(rad/s)], K_{iv} [Hz]), Coulomb level of the friction T_c [N m], median fit [%] of the position error and of the current, adequacy.

	change	axis	K_{pp}	K_{pv}	K_{iv}	T_c	error	current	adequate
I1	file gains, $K_{pp} = 2\pi$ `PL.KP` (WorkBench)	azimuth	62.8	250	1	0.54	-6	-146	no
		elevation	2.51	25	15	0.76	-10	-8	no
I2	file gains, $K_{pp} =$ `PL.KP` in 1/s	azimuth	10	250	1	0.54	71	-5	no
		elevation	0.4	25	15	0.76	-278	-25	no
I3	K_{pp}, K_{pv}, K_{iv} identified (grid + least squares), friction of the frequency fit	azimuth	9.72	125	0.25	0.54	84	5	no
		elevation	21	36	0.15	0.76	43	-198	no
I4	I3 + Coulomb level T_c identified, started from I3	azimuth	6.97	379	0.28	0.17	83	45	no
		elevation	17.3	29	0.127	0.10	42	-15	no
I5	grid on the gains with T_c of the moves, then gains and T_c together	azimuth	6.98	379	0.26	0.17	83	45	no
		elevation	6.78	53.8	2.88	0.38	81	68	yes

I1 → I2. Reading `PL.KP` in 1/s instead of 2π `PL.KP` brings the azimuth error fit from −6 to 71 %. The elevation becomes far too slow: its file value is `PL.KP` = 0.4.

I2 → I3. The gains that the files do not reproduce are identified within the AKD2G structure, from the best point of a grid, on the 1° sequences. The azimuth position gain comes out at 9.7 1/s, close to `PL.KP` = 10 in 1/s. The friction is still the one derived from the frequency fit (T_c 0.54 and 0.76 N m), which is too high for the moves, and the elevation current fit is −198 %.

I3 → I4. The Coulomb level is identified with the gains, starting from the rigid-body balance of the moves (0.17 and 0.34 N m). The azimuth current fit reaches 45 %. The elevation search started from the biased gains of I3 and ended in a local minimum.

I4 → I5. The grid is searched with the friction level of the moves already set, then gains and friction are fitted together. The elevation becomes adequate (81 % / 68 %). The azimuth stays at 83 % / 45 %: its current has an 8–10 Hz oscillation after the moves that does not depend on the fork damping or on the velocity gains (checked: 44.7–46.7 %).

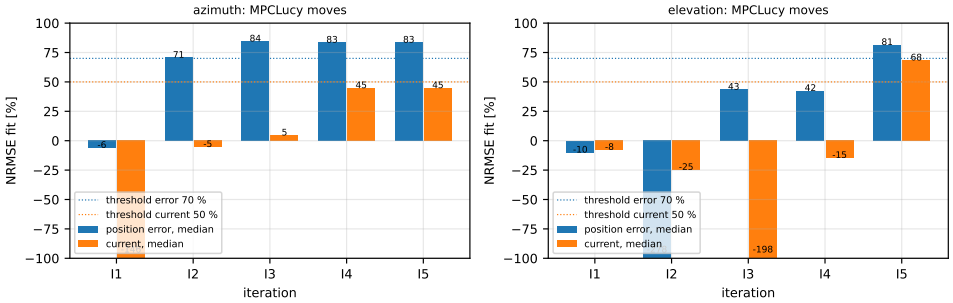


Figure 3: Median fits of the closed-loop iterations, with the adequacy thresholds.

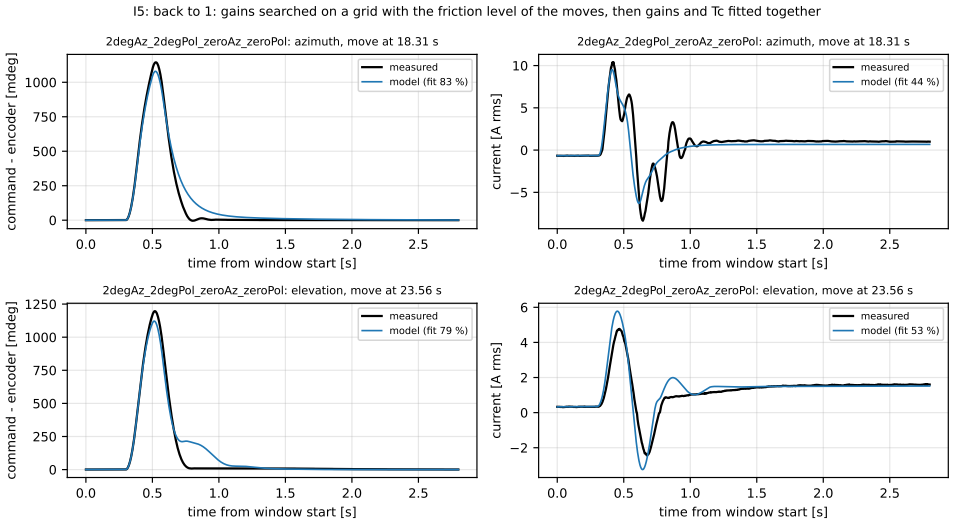


Figure 4: Validation moves of iteration I5 (2° sequences): position error and current, measured and model.

2.4 Imbalance of the telescope

Question. Could a strong uncertainty on the position of the centre of gravity explain the low-frequency part of the elevation Bode? A gravity moment gives a stiffness $m\,g\,d$ (with $m\,g = 150\,\mathrm{N}$), so the 190–290 N m/rad needed by the Bode would require d of 1.25–1.95 m.

Answer. No:

- The CAD offset (73 mm along the telescope axis) gives a static moment of 11 N m but only 0.78 N m/rad of stiffness at 0°.
- The holding torques of the sequences give a static moment of $-0.25\,\mathrm{N\,m}$.
- A conservative spring cannot produce the measured phase (Figure 5).

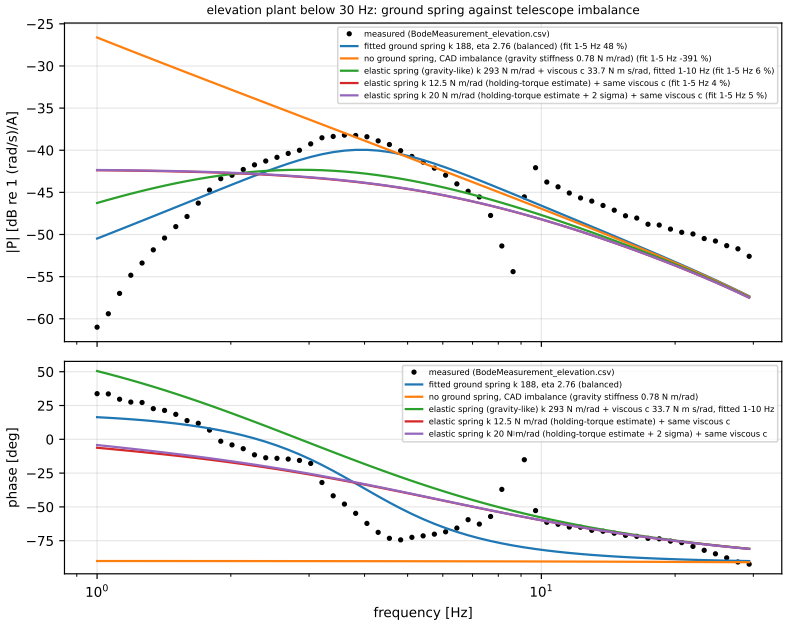


Figure 5: Elevation plant below 30 Hz: hysteretic ground spring against elastic springs of the size an imbalance could give.

3 Part B: plant driven by the measured current

3.1 Method

Model. The measured motor currents, low-passed at 150 Hz with zero phase, drive the plant, and the drives are not in the loop:

$$\tau_m = \kappa_i \, K_t \, i_{meas}(t) - b_m \, \omega_m. \tag{3.1}$$

The only goodness measure is the NRMSE fit of the encoder position of the moving axis. Each window runs from 0.3 s before a move to 0.5 s after the end of its command. This approach tests the mechanics and the friction alone, independently of the controller questions of Part A.

Elements needed with respect to Part A. With the current as input, the I5 model drifted at rest.

- A break-away (stiction) level. The friction is a Dahl element with the Stribeck curve of the LuGre model,

$$\dot{T}_f = \sigma_0 \Big(\dot{q} - |\dot{q}|_\varepsilon \, \frac{T_f}{g(\dot{q})} \Big), \qquad g(v) = T_c + (T_s - T_c)\, e^{-(v/v_s)^2}, \tag{3.2}$$

  which can hold up to T_s at rest and slides at T_c.

- A torque scale κ_i per axis, for the scale of the current channel, the $\pm 10\,\%$ tolerance of K_t and the CAD inertia.

- An initial state built from the static balance: holding torque, gravity when present, extra torques and twist of the shafts.

Fitting. The first guess is I5. For each axis, the parameters are screened one at a time by $\pm 30\,\%$ and kept when the cost changes by more than $2\,\%$. The least squares start from the best point of a small grid.

What the screening shows. The position histories do not see k_b, κ_r and the structural dampers (Table 6), which therefore keep the values of the frequency fit.

Table 6: Screening of the first run: relative change of the identification cost [%] for ±30 % of each parameter.

axis	parameter	cost change [%]	fitted
azimuth	κ_i	725.4	yes
azimuth	σ_0	7.4	yes
azimuth	T_c	217.0	yes
azimuth	T_s/T_c	16.7	yes
azimuth	v_s	1.0	no
azimuth	$c_{\psi,1}$	0.0	no
azimuth	$c_{\psi,2}$	0.0	no
elevation	κ_i	752.2	yes
elevation	σ_0	16.5	yes
elevation	T_c	846.0	yes
elevation	T_s/T_c	94.3	yes
elevation	v_s	5.1	yes
elevation	k_b	1.1	no
elevation	κ_r	0.3	no
elevation	$c_{\theta,1}$	0.0	no
elevation	$c_{\theta,2}$	0.1	no
elevation	$c_{\theta,3}$	0.0	no
elevation	$c_{\theta,4}$	0.3	no

3.2 Runs and what changed

First run. Torque scale and friction (σ_0, T_c, T_s/T_c, v_s), from I5.

- Azimuth adequate: validation 94.5 %.
- Elevation at 81.0 %, with an asymmetric error: the positive moves stop short of the target.

I6. The asymmetry looked like a constant torque, so gravity is switched on with a free centre-of-gravity offset d_x along the telescope axis.

- d_x comes out at -0.27 mm, a static moment of 0.04 N m: the imbalance does not explain the asymmetry.
- Elevation validation 86.8 %; the break-away ratio reaches its bound.

I7. The parameters identified on 1° moves left the 2° moves short, which suggests a speed-dependent friction. A viscous damping of the elevation bearing towards the fork is added ($b_{\theta,h}$, the $\sigma_2 v$ term of LuGre), and the bound of the break-away ratio is widened.

- Identification 95.9 %, validation 86.2 %: no gain.

I8 (requested after the steady-state offsets of I7):

- Azimuth: a two-level hysteresis (relay hysteron) on the bearing, $-Hs$ with $\dot{s} = (\dot{q} - |\dot{q}|_\varepsilon s)/\delta$. Its amplitude H goes to 0.006 N m, so it has no effect.
- Elevation: a torque $a \sin\theta + b \cos\theta + c$ of the absolute angle replaces the torsional (pre-sliding) spring, whose stiffness σ_0 is fixed stiff. It comes out as a spring of 18 N m/rad about a fixed neutral angle of 0.65°. Identification 96.3 %, validation 62.6 %: at 2° the torque exceeds the break-away level and the model slides before the move, while the axis stays at rest.

I9a. An elastic torque of the cables, $-K_c(\theta - \theta_0)$, is added on both axes with the friction of I7. The neutral angle θ_0 is set by the layout of the cables and is an unknown, here one per axis.

- Identification 95–97 %, validation 78.7 % (azimuth) and 71.5 % (elevation): overfitting.

I9b. As I9a, with θ_0 one unknown per sequence, since the cables may settle differently from one test to the next. It is fitted with the other parameters on the identification sequences, and estimated alone on each validation sequence with all other parameters fixed: an unknown initial condition, as `compare(...,'InitialCondition','estimate')`. The first estimate stayed at its start (0°); it was redone with a grid (correction in the summary of the run).

- Validation 94.9 % (azimuth) and 87.0 % (elevation).
- Azimuth: $K_c = 14.8$ N m/rad with θ_0 of 0.5–0.9°, consistent between the tests.

- Elevation: $K_c = 0.83\,\mathrm{N\,m/rad}$ with θ_0 of 3.3–5.7°, i.e. a nearly constant torque of 0.05–0.08 N m.

Table 7: Position fit [%] of the current-driven runs, median [min, max] over 8 (azimuth) and 10 (elevation) identification moves and 4 + 4 validation moves.

run	azimuth, identification	azimuth, validation	elevation, identification	elevation, validation
first run	96.2 [91, 98]	94.5 [91, 98]	86.1 [81, 94]	81.0 [58, 95]
I6	96.2 [91, 98]	94.5 [91, 98]	94.5 [88, 96]	86.8 [75, 91]
I7	96.2 [91, 98]	94.5 [91, 98]	95.9 [92, 97]	86.2 [80, 99]
I8	96.1 [91, 98]	94.4 [91, 98]	96.3 [93, 99]	62.6 [-0, 88]
I9a	95.4 [94, 99]	78.7 [77, 80]	96.5 [94, 98]	71.5 [61, 85]
I9b	98.5 [96, 99]	94.9 [94, 95]	97.2 [95, 98]	87.0 [81, 88]

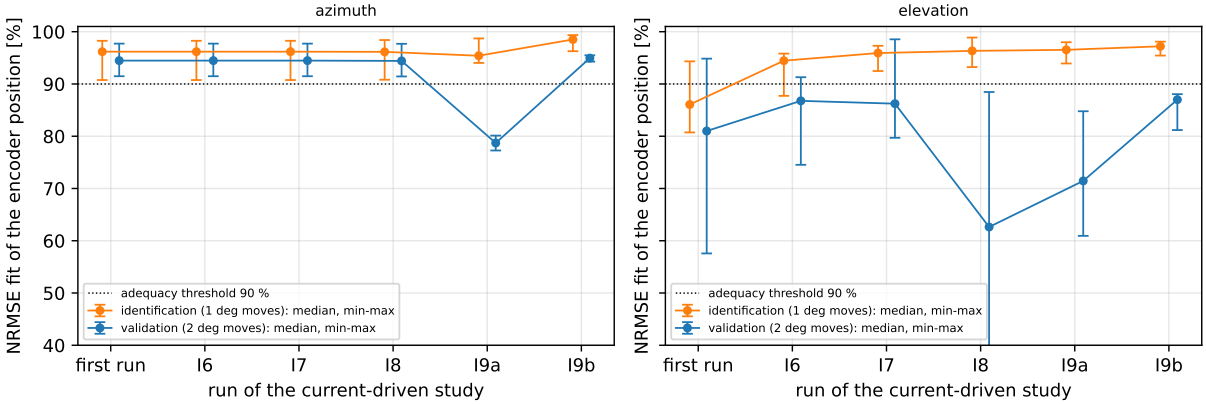


Figure 6: Position fit of the current-driven runs, identification and validation, with the adequacy threshold.

Table 8: Azimuth parameters of the current-driven runs (bold: fitted in that run, otherwise carried over or fixed). I9b: θ_0 of the two identification sequences, then of the two validation sequences.

parameter	unit	first run	I6	I7	I8	I9a	I9b
κ_i	-	**0.9062**	0.9062	0.9062	**0.9051**	**0.8562**	**0.8998**
σ_0	N m/rad	**202.8**	202.8	202.8	**191.4**	**401.5**	**1697**
T_c	N m	**0.2279**	0.2279	0.2279	**0.2212**	**0.1634**	**0.1805**
T_s/T_c	-	**1.477**	1.477	1.477	**2.593**	**2.22**	**1.016**
v_s	rad/s	0.001	0.001	0.001	0.001	0.001	0.001
d_x	mm	–	0	0	0	0	0
H	N m	–	–	–	**0.006057**	–	–
δ	rad	–	–	–	**0.0003705**	–	–
K_c	N m/rad	–	–	–	–	**16.52**	**14.8** 0.60 0.47 0.91
θ_0	deg	–	–	–	–	**0.5289**	0.90

Table 9: Elevation parameters of the current-driven runs, as Table 8.

parameter	unit	first run	I6	I7	I8	I9a	I9b
κ_i	-	**0.9374**	**0.9268**	**0.9739**	**0.9835**	**0.9354**	**1.027**
σ_0	N m/rad	**413.3**	**620.4**	**1202**	2e4	**1855**	**1766**
T_c	N m	**0.3959**	**0.3277**	**0.2594**	**0.2716**	**0.2422**	**0.2609**
T_s/T_c	-	**5.196**	**9.988**	**6.453**	**1.023**	**3.821**	**9.202**
v_s	rad/s	**0.006637**	0.006637	**0.01817**	0.01817	**0.01129**	**0.01417**
d_x	mm	–	**-0.2687**	**-0.3254**	0	0	0
$b_{\theta,h}$	N m s/rad	–	–	**0.03995**	0.03995	**0.3041**	**0.03586**
a	N m	–	–	–	**-18.4**	–	–
b	N m	–	–	–	**-0.01053**	–	–
c	N m	–	–	–	**0.2227**	–	–
K_c	N m/rad	–	–	–	–	**12.15**	**0.8335** 4.48 3.29 5.68
θ_0	deg	–	–	–	–	**0.7252**	3.53

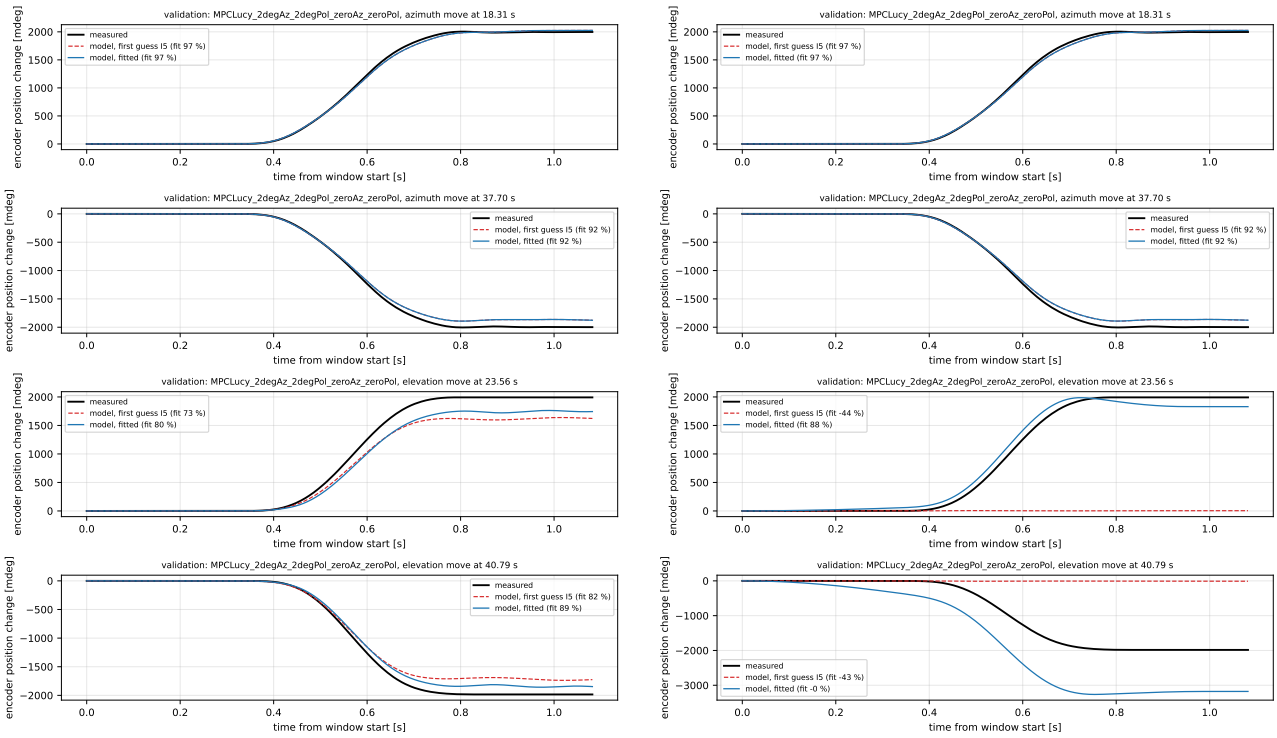


Figure 7: Validation moves (2° sequences), encoder position measured and simulated with the first guess and the fitted parameters: I7 (left) and I8 (right).

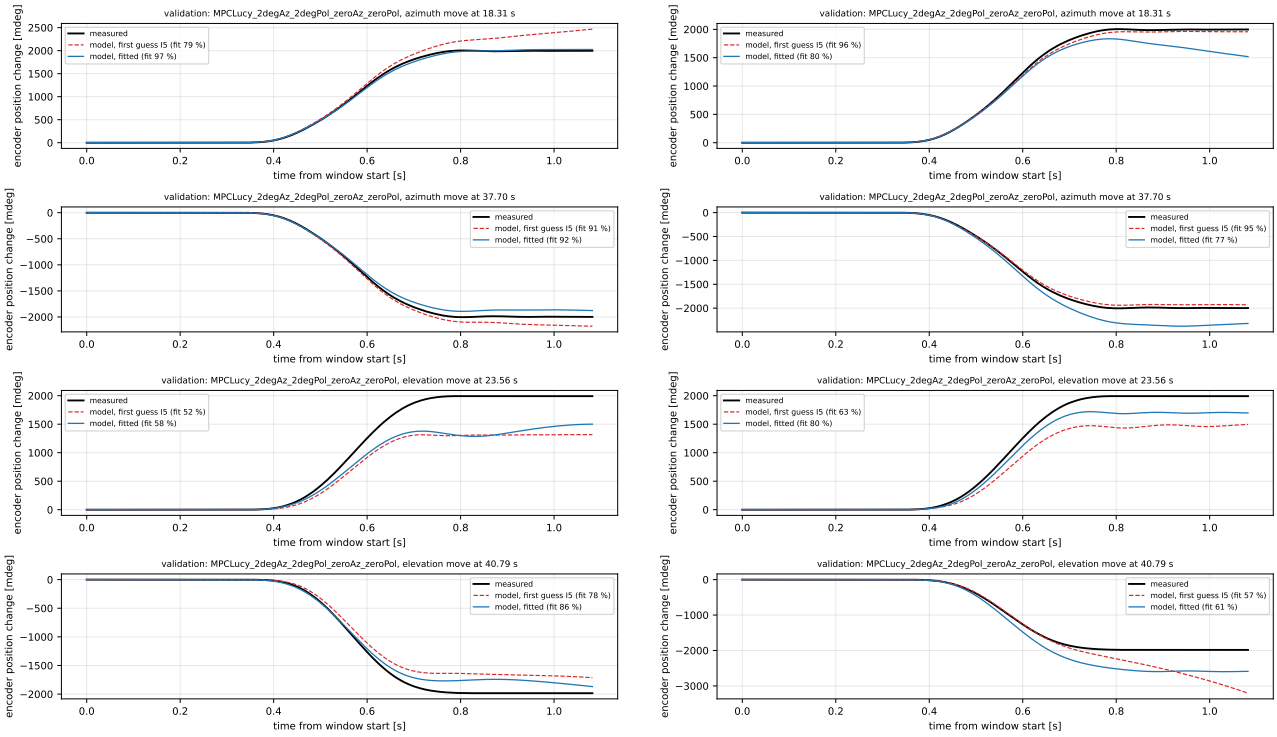


Figure 8: Validation moves of the first run (left) and of I9a (right).

4 Role of the parameters

Table 10: Parameters of the current-driven model: meaning and what in the position histories determines them.

parameter	unit	role
κ_i	–	Torque per measured current over K_t: with the CAD inertia it sets the acceleration of the rigid axis, hence the amplitude of the move. It is the best determined parameter.
σ_0	N m/rad	Pre-sliding stiffness of the bearing: elastic micro-motion before sliding, and the history-dependent "spring" $T_{f,0} + \sigma_0(\theta - \theta_{start})$ whose neutral point resets at every slide. It sets the small creep and recoil at the end of a move.
T_c	N m	Coulomb (sliding) friction: dissipation during the move, hence the shortfall of the travel.
T_s/T_c, v_s	–, rad/s	Break-away level and speed below which it applies: they decide whether the axis sticks at rest under the holding current and when it starts moving.
$b_{\theta,h}$	N m s/rad	Viscous friction of the elevation bearing; on moves of equal speed it is indistinguishable from T_c.
d_x	mm	Centre of gravity of the telescope along its axis: static moment $mgd_x \cos\theta$.
H, δ	N m, rad	Two-level hysteresis: torque $\pm H$ set by the direction of the last move, switching after a few δ.
a, b, c	N m	Torque of the absolute angle: over 0–2° only a (a stiffness) and $b + c$ (a constant) are separable.
K_c, θ_0	N m/rad, deg	Cable spring and its neutral angle: over a small range of angles the product $K_c\theta_0$ is a constant torque and the two are nearly collinear.
k_b, κ_r, c_*		Structural stiffness and dampers above 50 Hz: determined by the frequency responses, not seen by the position histories.

5 Identifiability analysis

Method (`identifiability.py`, `results_identifiability`). At the fitted point of the first run (azimuth), I7 (elevation) and I9a (both axes), the Jacobian J of the normalised position residuals is computed on the identification moves, by central differences in the coordinates of the fit ($\ln p$ for positive parameters, p for signed ones). From it the analysis derives:

- the condition number of J, with its columns scaled to unit norm;
- the covariance of the estimates, $\sigma^2(J^T J)^{-1}$, from which follow the correlations and the standard errors.

Consecutive samples of a smooth position history are strongly correlated. The information is therefore scaled by the effective number of independent samples, estimated from the autocorrelation of the residuals of each move. The standard errors come from a linearisation and should be read as orders of magnitude; the correlations are more robust.

Table 11: Identifiability at the fitted points: condition number, effective independent samples, fitted value, standard error (relative [%] for ln-parameters, absolute for signed ones) and correlations with $|r| > 0.9$.

model	axis	cond.	n_{eff}	parameter	value	st. error	strong correlations
first run (no cables)	azimuth	14.4	27	κ_i	0.9062	47	–
				σ_0	202.8	587	–
				T_c	0.2279	112	–
				T_s/T_c	1.477	8812	–
I7 (no cables)	elevation	224	64	κ_i	0.9739	24	–
				σ_0	1202	94	–
				T_c	0.2594	388	$b_{\theta,h}$ (-0.99)
				T_s/T_c	6.453	514	–
				v_s	0.01817	380	–
				d_x	-0.3254	0.12 (abs.)	–
				$b_{\theta,h}$	0.03995	28908	T_c (-0.99)
I9a (cable spring, one θ_0)	azimuth	17.5	35	κ_i	0.8562	35	T_c (+0.90)
				σ_0	401.5	2106	K_c (+0.92)
				T_c	0.1634	173	κ_i (+0.90)
				T_s/T_c	2.22	47600	–
				K_c	16.52	737	σ_0 (+0.92)
				θ_0	0.5289	0.22 (abs.)	–
I9a (cable spring, one θ_0)	elevation	215	41	κ_i	0.9354	69	–
				σ_0	1855	237	–
				T_c	0.2422	583	$b_{\theta,h}$ (-0.99)
				T_s/T_c	3.821	1337	–
				v_s	0.01129	1273	–
				$b_{\theta,h}$	0.3041	5384	T_c (-0.99)
				K_c	12.15	596	θ_0 (-1.00)
				θ_0	0.7252	1.4 (abs.)	K_c (-1.00)

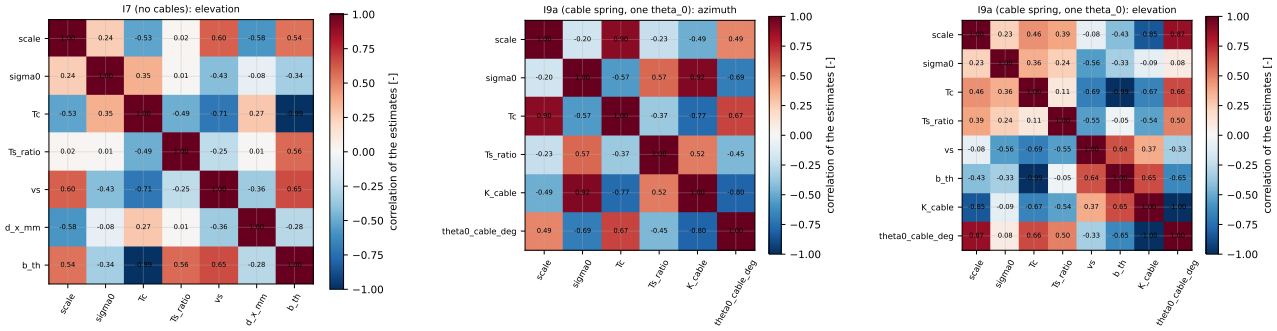


Figure 9: Correlation matrices of the estimates: I7 elevation, I9a azimuth, I9a elevation.

What it shows.

1. **The data hold little independent information.** Out of about 4000–4900 samples per axis, the 8–10 identification moves give only 27–64 effectively independent ones, 3–6 per move. All the moves are moreover alike: 1° at 10 deg/s, between 0 and 1°.

2. **Only the torque scale is reasonably determined** (standard error 24–69 %). The friction parameters have standard errors of hundreds or thousands of per cent.

3. **Strong correlations reveal parameters the data cannot separate:**

   - T_c with $b_{\theta,h}$ (-0.99): Coulomb and viscous friction on moves of equal speed;
   - K_c with θ_0 (-1.00, elevation): the cable spring is only a constant torque over 0–1°;
   - σ_0 with K_c ($+0.92$, azimuth): the pre-sliding and the cable stiffness exchange.

4. **The overfitting grows with the parameters.** The gap between identification and validation is:

   - azimuth: 1.7 points with 4 parameters, 16.7 with 6;
   - elevation: 9.7 points with 7 parameters, 25 with 8.

Note. Re-identifying more parameters on the same data would not help. Those left out (k_b, κ_r, the structural dampers) change the cost by less than 1–2 %, and the ones already fitted are not separable. The limit is the information in the identification data, not the number of free parameters.

6 Next steps

1. **New, more varied dataset** (to be provided): moves of several amplitudes (e.g. 0.5–5°), speeds and accelerations, from different start angles and in both directions, with long rests and slow ramps. Different speeds separate T_c from $b_{\theta,h}$; different angles separate K_c from θ_0; long rests determine the break-away level.

2. **Cross-validation by sequence.** Identify on all sequences but one, validate on the excluded one, in rotation. Choose the model structure (without cables; with cables; with or without viscous friction) by its cross-validation score, not by the identification fit.

3. **Fewer free parameters.** Fix or merge the parameters found not identifiable (T_s/T_c, v_s; $b_{\theta,h}$ with T_c), or give them priors; report every estimate with its uncertainty.

4. **Friction with memory at several scales** for the elevation: a Maxwell-slip model with a few elements, which represents the dependence on amplitude and history that a single Dahl element misses. Choose it by cross-validation.

5. **Open points of the closed loop:**

   - read the gains active in the drives during a test, since the elevation behaves as $K_{pp} \approx 7\,1/\mathrm{s}$ while the file has `PL.KP` $= 0.4$;
   - the 8–10 Hz current oscillation of the azimuth, possibly the compliance of the base or of the stator;
   - the motor-side inertia indicated by the 95 and 470 Hz resonances.

A Configurations and results

Table 12: Configuration files and results (folder `sim/gimbal_multidof/closedLoopModel`).

configuration	results
`config_vcl_study.yaml` (Part A)	`results_vcl_study/` and `config_vcl_fit.yaml` (fitted plant); `results_vcl_study/imbalance/`
`config_current_input.yaml`	`results_current_input/` (first run)
`config_current_input_I6.yaml`, `_I7`, `_I8`, `_I9a`, `_I9b`	`results_current_input_I6/` … `_I9b/` (I9b: `validation_corrected.json`, `theta0_validation_grid.json`)
`identifiability.py`	`results_identifiability/`